SAGAR PUBLIC SCHOOL

Integrated Programme • Topic: Satellite Motion

PRACTICE SHEET – 2

Satellite Motion: Orbital Speed, Period, Energy and Orbits

Name: _____	Class / Section: _____	Date: _____
Time Allowed: 60 Mins	**Total Questions: 15**	Roll No.: _____

Instructions: Attempt all questions. Section A has only one correct option; Section B may have more than one correct option; Section C is match the following; Section D needs full working. Time Allowed: 60 Mins.

Section A – Single Correct Option (Q.1 to Q.6)

Q.1 A spherical uniform planet is rotating about its axis. The velocity of a point on its equator is V. Due to the rotation of planet about its axis the acceleration due to gravity g at equator is $1/2$ of g at poles. The escape velocity of a particle on the pole of planet in terms of V is

(A) $V_e = 2V$ (B) $V_e = V$
(C) $V_e = V/2$ (D) $V_e = \sqrt{3}\,V$

[Motion, Ex-I Q.8]

Q.2 Two planets A and B have the same material density. If the radius of A is twice that of B, then the ratio of the escape velocity $\dfrac{v_A}{v_B}$ is

(A) 2 (B) $\sqrt{2}$ (C) $1/\sqrt{2}$ (D) $1/2$

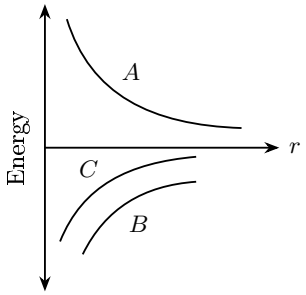
[Motion, Ex-I Q.9]

Q.3 Two identical satellites are at the heights R and $7R$ from the Earth's surface. Then which of the following statement is incorrect. (R = radius of the Earth)

(A) Ratio of total energy of both is 5
(B) Ratio of kinetic energy of both is 4
(C) Ratio of potential energy of both 4
(D) Ratio of total energy of both is 4 and ratio of magnitude of potential to kinetic energy is 2

[Nucleus, O-1 Q.14]

Q.4 The figure shows the variation of energy with the orbit radius of a body in circular planetary motion. Find the correct statement about the curves A, B and C



(A) A shows the kinetic energy, B the total energy and C the potential energy of the system.
(B) C shows the total energy, B the kinetic energy and A the potential energy of the system.
(C) C and A are kinetic and potential energies respectively and B is the total energy of the system.
(D) A and B are kinetic and potential energies and C is the total energy of the system.

[Nucleus, O-1 Q.17]

Q.5 The energy required to take a satellite to a height 'h' above Earth surface (radius of Earth $= 6.4 \times 10^3$ km) is E_1 and kinetic energy required for the satellite to be in a circular orbit at this height is E_2. The value of h for which E_1 and E_2 are equal, is:

(1) 1.28×10^4 km (2) 1.6×10^3 km
(3) 6.4×10^3 km (4) 3.2×10^3 km

[Nucleus, JM Q.11 [JEE Main-2019]]

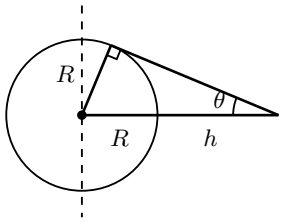
Q.6 A satellite is in an elliptical orbit around a planet P. It is observed that the velocity of the satellite when it is farthest from the planet is 6 times less than that when it is closest to the planet. The ratio of distances between the satellite and the planet at closest and farthest points is :

(1) 1 : 6 (2) 3 : 4 (3) 1 : 3 (4) 1 : 2

[Nucleus, JM Q.29 [JEE Main-2020]]

Section B – One or More Than One Option Correct (Q.7)

Q.7 A geostationary satellite is at a height h above the surface of earth. If earth radius is R



(A) The minimum colatitude on earth upto which the satellite can be used for communication is $\sin^{-1}\left(\frac{R}{R+h}\right)$

(B) The maximum colatitudes on earth upto which the satellite can be used for communication is $\sin^{-1}\left(\frac{R}{R+h}\right)$

(C) The area on earth escaped from this satellite is given as $2\pi R^2(1+\sin\theta)$

(D) The area on earth escaped from this satellite is given as $2\pi R^2(1+\cos\theta)$

[Motion, Ex-II Q.4]

Section C – Match the Following (Q.8 to Q.10)

Each entry of Column I may match one or more entries of Column II.

Q.8 A satellite is in a circular equatorial orbit of radius 7000 km around the Earth. If it is transferred to a circular orbit of double the radius

Column I	Column II
(A) Angular momentum	(P) Increases
(B) Area of Earth covered by satellite signal	(Q) Decreases
(C) Potential energy	(R) Becomes double
(D) Kinetic energy	(S) Becomes half

[Nucleus, O-2 Q.21]

Q.9 In elliptical orbit of a planet, as the planet moves from aphelion position to perihelion position,

Column I	Column II
(A) Speed of planet	(P) Remains same
(B) Distance of planet from centre of Sun	(Q) Decreases
(C) Potential energy	(R) Increases
(D) Angular momentum about centre of Sun	(S) Cannot say

[Nucleus, O-2 Q.22]

Q.10 Match Column I with Column II.

Column I	Column II
(A) Kinetic energy of a particle in gravitational field is increasing	(P) work done by gravitational force should be positive
(B) Potential energy of a particle in gravitational field is increasing	(Q) work done by external force should be non zero
(C) Mechanical energy of a particle in gravitational field is increasing	(R) work done by gravitational force should be negative
	(S) cannot say anything

[Nucleus, O-2 Q.23]

Section D – Subjective Problems (Q.11 to Q.15)

Q.11 A satellite is moving in a circular orbit around the earth. The total energy of the satellite is $E = -2\times10^5$ J. The amount of energy to be imparted to the satellite to transfer it to a circular orbit where its potential energy is $U = -2\times10^5$ J is equal to _____.

[Nucleus, S-1 Q.8]

Q.12 A satellite of mass m is orbiting the earth in a circular orbit of radius r. It starts losing energy due to small air resistance at the rate of C J/s. Then the time taken for the satellite to reach the earth is _____.

[Nucleus, S-1 Q.9]

Q.13 A satellite close to the earth is in orbit above the equator with a period of rotation of 1.5 hours. If it is above a point P on the equator at some time, it will be above P again after time _____.

[Nucleus, S-1 Q.10]

Q.14 A binary star has a period (T) of 2 earth years while distance L between its components having masses M_1 and M_2 is four astronomical units. If $M_1 = M_S$ where M_S is the mass of sun, find the ratio $M_2/5M_S$.

[Nucleus, S-1 Q.11]

Q.15 A pair of stars rotates about a common center of mass. One of the stars has a mass M which is twice as large as the mass m of the other. Their centres are at a distance d apart, d being large compared to the size of either star.

(a) Derive an expression for the period of rotation of the stars about their common centre of mass in terms of d, m, G.

(b) Compare the angular momentum of the two stars about their common centre of mass by calculating the ratio L_m/L_M.

(c) Compare the kinetic energies of the two stars by calculating the ratio K_m/K_M.

[Nucleus, S-1 Q.12]

ANSWER KEY – Practice Sheet 2

Answers are taken from the source answer keys (Motion: Answer Ex-I, Ex-II; Nucleus: Answer Key page, Exercises S-1, O-1, O-2, JM). JM answers are option numbers.

1. [Motion Ex-I Q.8] A

2. [Motion Ex-I Q.9] A

3. [Nucleus O-1 Q.14] A

4. [Nucleus O-1 Q.17] D

5. [Nucleus JM Q.11] (4)

6. [Nucleus JM Q.29] (1)

7. [Motion Ex-II Q.4] A, C

8. [Nucleus O-2 Q.21] (A)→P; (B)→P; (C)→P, S; (D)→Q, S

9. [Nucleus O-2 Q.22] (A)→R; (B)→Q; (C)→Q; (D)→P

10. [Nucleus O-2 Q.23] (A)→S; (B)→R; (C)→Q

11. [Nucleus S-1 Q.8] 1×10^5 J

12. [Nucleus S-1 Q.9] $t = \dfrac{GMm}{2C}\left(\dfrac{1}{R_e} - \dfrac{1}{r}\right)$

13. [Nucleus S-1 Q.10] 1.6 hours if it is rotating from west to east, 24/17 hours if it is rotating from west to east (printed identically in the source key)

14. [Nucleus S-1 Q.11] 3

15. [Nucleus S-1 Q.12] (a) $T = \dfrac{2\pi d^{3/2}}{\sqrt{3Gm}}$; (b) 2; (c) 2

Q.1 and Q.2 repeat Nucleus O-1 Q.10 and Q.13 (Sheet 1, Q.3 and Q.5); both appear in the topic sheet because the mapping lists them.

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SELF ASSESSMENT – Practice Sheet 2

Fill this in after checking your answers with the Answer Key.

Time Allowed: 60 Mins	**Time Taken:** _____ Mins
Start time: _____	End time: _____
Finished within the allowed time? ☐ Yes ☐ No	(extra / saved time: _____ Mins)

Total	Attempted	Correct	Incorrect	Not attempted
15				

Score (Correct / 15): _____ Percentage: _____ %

Question-wise tracker write ✓ (correct), ✗ (incorrect) or – (not attempted)

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	

Reason for the incorrect / unattempted questions (write question numbers)

Concept not clear: _____	Formula forgotten: _____
Calculation / silly mistake: _____	Ran out of time: _____

Personal feedback guide (tick the band that matches your score)

	Correct	What to do next
☐	14–15	Excellent. Concept and speed are both strong. Move to harder JEE-level problems and keep revising formulae once a week.
☐	11–13	Good. Re-do every incorrect question without looking at the solution, then note the formula or idea you missed.
☐	7–10	Fair. Revise the theory of escape speed, orbital speed, energies of a satellite and Kepler's laws, then re-attempt this sheet after 3 days.
☐	0–6	Needs attention. Re-read the chapter notes and solved examples first, then attempt this sheet again in two parts of 30 mins.

My personal feedback / action plan:

_____

_____

_____

_____

Teacher's remarks:

_____

_____

Student's signature: _____ Teacher's signature: _____